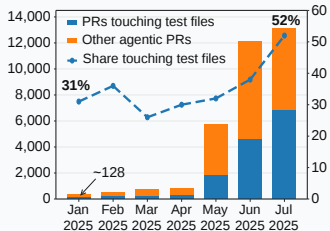


1. Real-world motivation

Test count explosion



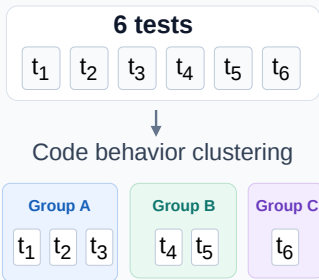
**Human review
budget is limited**



**Feedback / triage
cost grows**

2. Count $\neq$ diversity

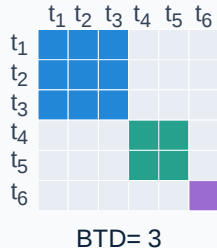
Multiple tests may test one code behavior



**Fix one code behavior
might fix multiple tests**

3. BTD measures distinct code behaviors

Use repair transfer to quantify how tests relate.



**Repair transfer measures
distinct code behavior**

4. Optimize test suite

Prune redundant tests and expand with new, diverse tests.

Pruning

Select from each group

t_1 t_2 t_3 t_4 t_5 t_6

t_1 t_4 t_6

Count: 6 $\rightarrow$ 3

BTD: 3 $\rightarrow$ 3

Fewer tests to review and execute

Expansion

Testing new code behaviors

t_1 t_2 t_3 t_4 t_5 t_6 + t_7

t_1 t_2 t_3 t_4 t_5 t_6 t_7

Count: 6 $\rightarrow$ 7

BTD: 3 $\rightarrow$ 4

Prefer tests that test new behaviors.